

Jaejin Cha

AI Systems | Cloud & Data Platforms | Robotics & Autonomous Systems
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SUMMARY

MS Computer Science student at Texas A&M, graduating December 2026, and Software Engineer at HPE building enterprise AI applications for natural-language data exploration, RAG, tool use, and database-backed workflows on Oracle and OCI. Research spans multi-robot communication and graph-based planning, autonomous-driving simulation, reinforcement and imitation learning, and clinical data modeling.

TECHNICAL SKILLS

Languages: Python, C++, SQL, PL/SQL, JavaScript; **AI/ML:** PyTorch, scikit-learn, OpenCV, reinforcement learning, imitation learning
LLM: RAG, tool calling, text-to-SQL, LangChain; **Systems:** Linux, Docker, OCI, Oracle Database 19c/26ai, Oracle APEX, REST APIs, Git

EXPERIENCE

- Hewlett Packard Enterprise** Spring, TX
Software Engineer – AI Applications and Enterprise Systems Intern: May 2025 – May 2026 | Full-time: Jun 2026 – Present
- Built internal AI applications enabling operations leaders to query operational data in natural language and generate on-demand insights.
 - Implemented text-to-SQL and tool-calling workflows with Oracle AI Database, PL/SQL packages, and business-rule guardrails.
 - Built a custom vector store and similarity-search layer in Oracle 19c using PL/SQL, REST APIs and embedding models.
 - Developed Oracle APEX interfaces with persistent chat history, session management, and OCI-integrated AI workflows.
- NIH AIM-AHEAD Collaborative Research** Remote
Research Assistant - Texas A&M, University of Washington, and Emory University Feb 2025 – May 2026
- Built cardiovascular EHR data pipelines for cohort construction, feature engineering, longitudinal data preparation, and survival modeling.
 - Evaluated survival models using calibration, performance metrics, and subgroup fairness analysis on real-world clinical data.
- ENDEAVR Institute** College Station, TX
Machine Learning Engineer Intern Oct 2024 – Feb 2025
- Developed autonomous-driving imitation-learning components using GAIL and PPO, including generator-side implementation and simulation-oriented training workflows.
- University of Delaware** Newark, DE
Summer Research Intern Jun 2023 – Aug 2023
- Implemented DQN and DDPG models in an autonomous-driving simulator for trajectory control and energy-efficiency experiments.

RESEARCH & PUBLICATIONS

- Jaejin Cha, Dylan Shell.** “Planning and Execution for Multi-Robot Cyclic Rendezvous via Graphs of Convex Sets”. IROS (accepted). 2026
- Developed graph-based planning and execution methods for communication-aware coordination among robots following cyclic paths.
- Jaejin Cha.** “Rendezvous-Based Multi-Robot Communication Under Cyclic Paths”. Master’s thesis (defended). 2026

SELECTED PROJECTS

- NatureNet** | Python, PyTorch, OpenCV, Flask, React.js, Node.js, Express.js 2024
- Built a full-stack wildlife detection and alerting system using a custom CV model, inference API, and SMS/email notification workflows.
- JAL-AM Trading Simulator** | Python, PyTorch, NumPy 2024 – 2025
- Formulated a market simulator as an MDP and implemented Joint-Action Learning with Agent Modeling for decision-making experiments.
- MAES Points Reward System** | Ruby on Rails, PostgreSQL, Heroku 2024
- Built and deployed a points-based rewards platform for a Texas A&M student organization using Ruby on Rails, PostgreSQL, and Heroku.
 - Contributed to requirements analysis, system design, implementation, testing, deployment, and user training within an Agile team.

EDUCATION

- Texas A&M University** College Station, TX
M.S. Computer Science, GPA: 3.71/4.00 - Thesis in Multi-Robot Systems (Advisor: Dr. Dylan Shell) Expected Dec 2026
- Texas A&M University** College Station, TX
B.S. Computer Science, Minor in Mathematics Aug 2024

LEADERSHIP & ACTIVITIES

- Texas Aimbots** | Computer Vision Developer Sep 2023 – May 2024
- Implemented OpenCV-based target detection and contour extraction for RoboMaster University League competition robots.
- Aggie Coding Club** | Project Manager Feb 2022 – May 2022
- Led 20+ members in web-development workshops and coordinated React projects including personal websites and e-commerce interfaces.